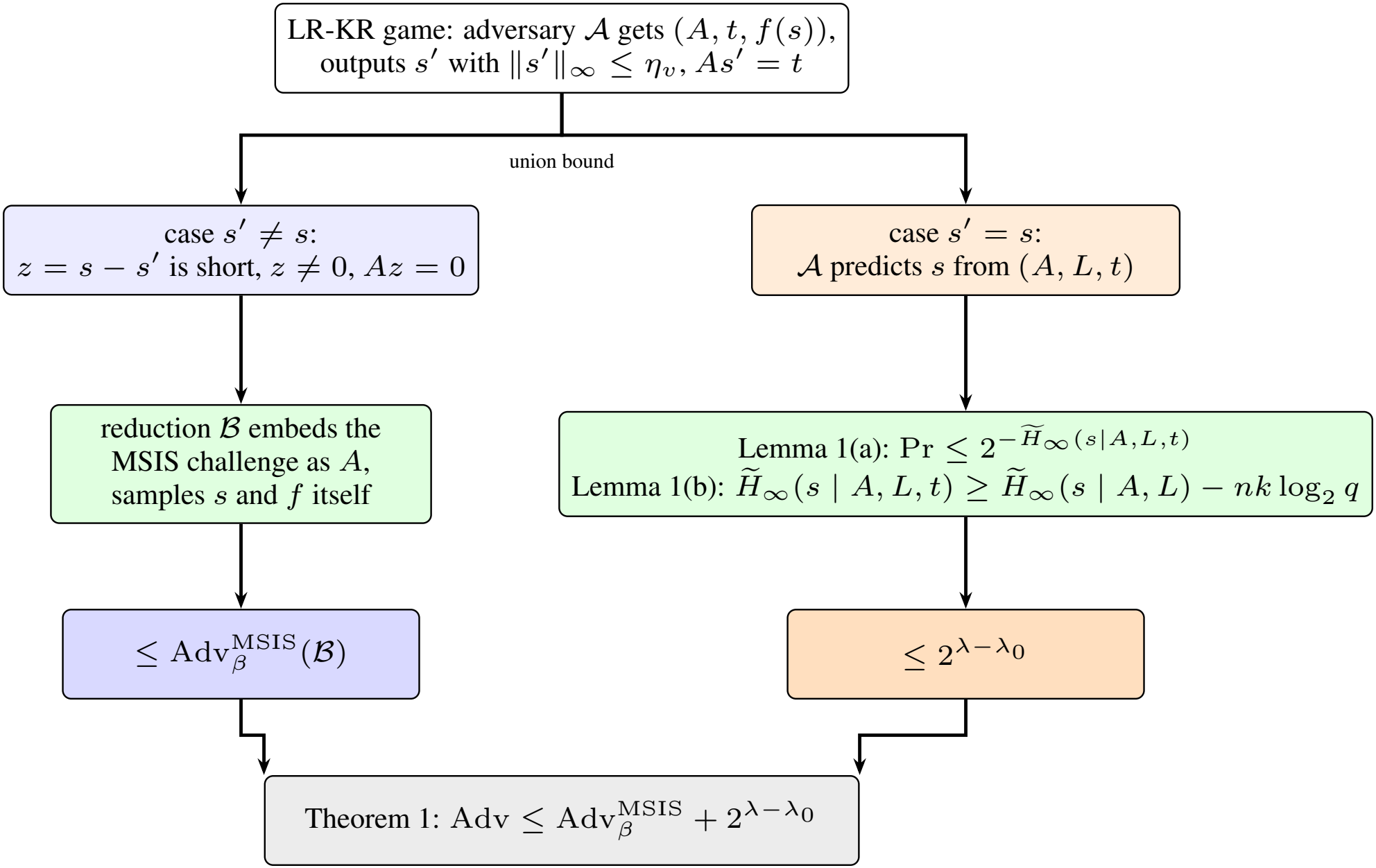

LR-KR game: adversary $\mathcal{A}$ gets $(A, t, f(s))$, outputs s' with $\|s'\|_\infty \le \eta_v$, $As' = t$
union bound
case $s' \neq s$: $z = s - s'$ is short, $z \neq 0$, $Az = 0$
case $s' = s$: $\mathcal{A}$ predicts s from (A, L, t)
reduction $\mathcal{B}$ embeds the MSIS challenge as A, samples s and f itself
Lemma 1(a): $\Pr \le 2^{-\widetilde{H}_\infty(s|A,L,t)}$
Lemma 1(b): $\widetilde{H}_\infty(s \mid A, L, t) \ge \widetilde{H}_\infty(s \mid A, L) - nk \log_2 q$
$\le \mathrm{Adv}_\beta^{\mathrm{MSIS}}(\mathcal{B})$
$\le 2^{\lambda - \lambda_0}$
Theorem 1: $\mathrm{Adv} \le \mathrm{Adv}_\beta^{\mathrm{MSIS}} + 2^{\lambda - \lambda_0}$